Worksheet 7.2 — Combining Functions

Grade 12 Advanced Functions (MHF4U) · Unit 7: Rates of Change and Combining Functions

The Big Ideas

Combine functions point by point: $(f \pm g)(x) = f(x) \pm g(x)$, $(fg)(x) = f(x)g(x)$, $\left(\frac{f}{g}\right)(x) = \dfrac{f(x)}{g(x)}$ (with $g \neq 0$).

- $(f \pm g)(x) = f(x) \pm g(x)$; $(fg)(x) = f(x)g(x)$.
- Domain = intersection of the two domains.
- For $\frac{f}{g}$, also exclude $g(x) = 0$.

Core Concepts

Arithmetic of functions

Functions can be added, subtracted, multiplied, and divided point by point: $(f + g)(x) = f(x) + g(x)$, and so on.

Domain of the result

The combined function is defined only where *both* pieces are defined, and division also excludes where the divisor is zero.

Reading combined graphs

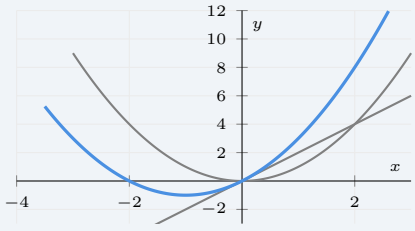
Add or subtract the y-values at each x to picture the sum or difference of two graphs.

Worked Examples

Example 1 — Sum

$f = x^2$, $g = 2x$. Find $(f + g)(x)$.

Solution. $x^2 + 2x$. Graphs add point by point:



Example 2 — Difference

Find $(f - g)(x)$.

Solution. $x^2 - 2x$.

Example 3 — Product

Find $(fg)(x)$.

Solution. $x^2 \cdot 2x = 2x^3$.

Example 4 — Quotient

Find $\left(\frac{f}{g}\right)(x)$ and its domain.

Solution. $\dfrac{x^2}{2x} = \dfrac{x}{2}$, with $x \neq 0$.

Example 5 — Evaluate

Find $(f+g)(2)$.

Solution. $f(2)+g(2) = 4+4 = 8$.

Example 6 — Sum

$f = x+4,\ g = x-4$. Find $(f+g)(x)$.

Solution. $2x$.

Example 7 — Product

Same f, g: find $(fg)(x)$.

Solution. $(x+4)(x-4) = x^2-16$.

Example 8 — Difference

$f = x^2,\ g = 5$. Find $(f-g)(x)$.

Solution. x^2-5.

Example 9 — Evaluate

$f = 2x,\ g = x-1$. Find $(f+g)(4)$.

Solution. $8+3 = 11$.

Your Turn — Show Your Work

For each question, show all steps clearly in the space provided.

Question 1

$f = x + 6,\ g = x - 2$. Find $(f + g)(x)$.

Question 2

Same f, g: find $(fg)(x)$.

Question 3

$f = x^2,\ g = 7$. Find $(f - g)(x)$.

Question 4

$f = 3x,\ g = x + 1$. Find $(f + g)(2)$.

Question 5

$f = x^3,\ g = x - 2$. Find the domain of $\frac{f}{g}$.

Question 6

$f = 3x,\ g = x^2$. Find $(f+g)(x)$.

Question 7

$f = x^2,\ g = x+1$. Find $(fg)(x)$.

Question 8

$f = x+3,\ g = x-3$. Find $(fg)(x)$.

Question 9

$f = x^2,\ g = x+1$. Find $(f+g)(1)$.

Question 10

$f = 5,\ g = x$. Find $(f - g)(x)$.

Question 11

$f = x^3,\ g = x$. Find $\left(\frac{f}{g}\right)(x)$ (and restriction).

Question 12

$f = x,\ g = x$. Find $(fg)(x)$.

Question 13 — Challenge

$f = x^2 - 1,\ g = x + 1$. Find $\left(\frac{f}{g}\right)(x)$ and state the restriction.

Question Recap & Answer Key

Question Recap

1. $f = x + 6,\ g = x - 2$. Find $(f + g)(x)$.
2. Same f, g: find $(fg)(x)$.
3. $f = x^2,\ g = 7$. Find $(f - g)(x)$.
4. $f = 3x,\ g = x + 1$. Find $(f + g)(2)$.
5. $f = x^3,\ g = x - 2$. Find the domain of $\frac{f}{g}$.
6. $f = 3x,\ g = x^2$. Find $(f + g)(x)$.
7. $f = x^2,\ g = x + 1$. Find $(fg)(x)$.
8. $f = x + 3,\ g = x - 3$. Find $(fg)(x)$.
9. $f = x^2,\ g = x + 1$. Find $(f + g)(1)$.
10. $f = 5,\ g = x$. Find $(f - g)(x)$.
11. $f = x^3,\ g = x$. Find $\left(\frac{f}{g}\right)(x)$ (and restriction).
12. $f = x,\ g = x$. Find $(fg)(x)$.
13. $f = x^2 - 1,\ g = x + 1$. Find $\left(\frac{f}{g}\right)(x)$ and state the restriction.

✓ Answer Key

1. $2x + 4$
2. $x^2 + 4x - 12$
3. $x^2 - 7$
4. 9
5. $x \neq 2$
6. $x^2 + 3x$
7. $x^3 + x^2$
8. $x^2 - 9$
9. 3
10. $5 - x$
11. $x^2,\ x \neq 0$
12. x^2
13. $x - 1,\ x \neq -1$